

7 (a) Horizontal lines [B1]

For between $x = 0$ cm to $x = 5.0$ cm, V value is 380 V

For between $x = 40.0$ cm to $x = 50.0$ cm, V value is 750 V

(b) Electric force will be directed towards sphere A and decreasing in magnitude, from surface of A to position of lowest potential

At position of lowest potential, electric force is zero

Electric force will be directed towards sphere B and increasing in magnitude, from position of lowest potential to surface of sphere B.

([B1] describing how the magnitude and direction of force is changing from A to lowest potential.

[B1] identify point of lowest potential to be zero force/ neutral point

[B1] describing how the magnitude and direction of force is changing from lowest point to B.)

(c) Position of zero field strength/ neutral point/ minimum potential is closer to sphere A [M1] (or potential at surface of/ close to A is smaller than potential at surface of/ close to B)

Therefore sphere A has a smaller magnitude of charge [A1]

(d) Method 1: Consider a point of resultant potential

Consider $x = 30$ cm, $V = 400$ V (or any other appropriate point)

$$V_A + V_B = 400$$

$$\frac{1}{4\pi\epsilon_0} \left(\frac{1.2 \times 10^{-9}}{0.30} \right) + \frac{1}{4\pi\epsilon_0} \left(\frac{Q_B}{0.50 - 0.30} \right) = 400 \quad [\text{C1}]$$

$$Q_B = 8.1 \times 10^{-9} \text{ C} \quad [\text{A1}]$$

If consider $x = 5.0$ cm, $V = 380$ V, then $Q_B = 8.22 \times 10^{-9} \text{ C}$

If consider $x = 40.0$ cm, $V = 750$ V, then $Q_B = 8.04 \times 10^{-9} \text{ C}$

Method 2: Consider neutral point

$x = 14.0$ cm

$$|E_A| = |E_B|$$

$$\frac{1}{4\pi\epsilon_0} \left(\frac{1.2 \times 10^{-9}}{0.14^2} \right) = \frac{1}{4\pi\epsilon_0} \frac{Q_B}{(0.50 - 0.14)^2} \quad [\text{C1}]$$

$$Q_B = 7.9 \times 10^{-9} \text{ C} \quad [\text{A1}]$$

allow range of $7.19 \text{ nC} \leq Q_B \leq 8.77 \text{ nC}$

based on readoff of position of neutral point to be $13.5 \text{ cm} \leq Q_B \leq 14.5 \text{ cm}$

(e) Method 1

Kinetic energy of electron = $\frac{1}{2} m v^2$

$$= \frac{1}{2} (9.11 \times 10^{-31}) (9.7 \times 10^6)^2$$

$$= 4.3 \times 10^{-17} \text{ J} \quad [\text{M1}]$$

Gain in electric potential energy from surface of A to neutral point

$$= e (V_{\text{neutral point}} - V_{\text{surface of A}})$$

$$= -1.6 \times 10^{-19} (280 - 380)$$

$$= 1.6 \times 10^{-17} \text{ J} \quad [\text{M1}]$$

Since kinetic energy of electron is more than required gain in electric potential energy from surface of A to neutral point,

electron is able to reach the surface of B. [A1]

Method 2

Minimum kinetic energy required to reach B

= gain in electric potential energy from surface of A to neutral point

$$= e (V_{\text{neutral point}} - V_{\text{surface of A}})$$

$$= -1.6 \times 10^{-19} (280 - 380)$$

$$= 1.6 \times 10^{-17} \text{ J} \quad [\text{M1}]$$

$$\text{So minimum launch speed} = \sqrt{\frac{2(1.6 \times 10^{-17})}{9.11 \times 10^{-31}}} = 5.9 \times 10^6 \text{ m s}^{-1} \quad [\text{M1}]$$

Since electron is ejected at a higher speed than the minimum launch speed,

electron is able to reach the surface of B. [A1]

